Worksheet 8-9 to 8-12 B Volumes Disc & Washer #1
AP Calculus Name_____

1.) Find the volume of the solid formed when the region between $y = x$ and $y = 4x - x^2$ is revolved about the x-axis.

2.) What is the volume of the solid formed when the region bounded by $x = y^2$, $x = 0$, and $y = 3$ is revolved about the y-axis?

3.) A flower vase is formed by rotating the portion of the parabola $y^2 = 4(x - 1)$ between $y = 0$, $y = 2$, and $x = 0$ about the y-axis. What is the volume of the vase?

4.) Find the volume of the solid formed when the region bounded by $y = x^3$ and $y = 4x$ in the first quadrant is revolved around the y-axis.

5.) Find the volume of the solid formed when the region bounded by the line $y = x - 2$, the x-axis, and $x = 4$ is revolved about the line $x = -1$.

6.) Find the volume of the solid formed when the region between $x = y^2$ and $x = 4$ in the first quadrant is revolved about the line $x = -1$.

7.) What is the volume of the solid formed when the region between $y = x^2$ and $y = 1$ is revolved about the line $y = -2$?

8.) What is the volume of the solid formed by the region between the x-axis, $x = 4$, and $y = x^2$ when it is revolved about the line $x = 4$?

9.) Find the volume of the solid formed when the region between $y = x^2$, $x = 0$, and $y = 9$ is revolved about the line $x = 4$.

10.) The region bounded by $y = e^x$, $x = 0$, $x = 3$, and $y = 0$ is revolved about the line $y = -1$. What is the volume of the solid formed?

Answers:

1.) $\frac{108\pi}{5}$ 2.) $\frac{243\pi}{5}$ 3.) $\frac{56\pi}{15}$ 4.) $\frac{128\pi}{15}$ 5.) $\frac{52\pi}{3}$

6.) $\frac{544\pi}{15}$ 7.) $\frac{104\pi}{15}$ 8.) $\frac{128\pi}{3}$ 9.) $\frac{207\pi}{2}$ 10.) $\left(\frac{e^6}{2} + 2e^3 - \frac{5}{2}\right)\pi$